

Summing up the alg. results in.

$\hat{\theta}^*(0)$ necessary, as well as S_0 [initialization]
 for $k=1, 2, 3, \dots$ (with $\hat{\theta}^*(k-1)$ and S_{k-1}) available:

$$\begin{cases} \text{gain: } S_k = \mu S_{k-1} + \varphi(k) \varphi^T(k) \\ \text{prediction: } \begin{cases} \underline{y}(k) = S_k^{-1} \varphi(k) \\ \hat{y}(k | \hat{\theta}^*(k-1)) = \varphi^T(k) \hat{\theta}^*(k-1) \\ \hat{\theta}^*(k) = \hat{\theta}^*(k-1) + \underline{y}(k) [y(k) - \hat{y}(k | \hat{\theta}^*(k-1))] \end{cases} \end{cases}$$

At each time k we have to invert $S_k \Rightarrow S_k$ must be invertible.
 Moreover we should have $S_k > 0$ for our solution to make sense.
 Inversion could be computationally complex and subject to numerical errors. To avoid inversion we resort to the

MATRIX INVERSION LEMMA

Given any matrices F, G, H, K of suitable dimensions $\Rightarrow (F + G H K)^{-1} = F^{-1} - F^{-1} G (H^{-1} + K F^{-1} G)^{-1} K F^{-1}$

We choose :

$$F \leftarrow \mu S_{k-1}$$

$$G \leftarrow \varphi(k)$$

$$H \leftarrow 1$$

$$K \leftarrow \varphi^T(k)$$

Note: K is a matrix, k = time instant

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$$\Rightarrow F + G H K = \mu S_{k-1} + \varphi(k) \varphi^T(k) = S_k$$

Applying the matrix inversion lemma:

$$\Rightarrow S_k^{-1} = \left(\begin{matrix} \mu \\ S_{k-1} + \varphi(k) \cdot 1 \cdot \varphi^T(k) \end{matrix} \right)^{-1} = \frac{1}{\mu} S_{k-1}^{-1} \frac{1}{\mu} S_{k-1}^{-1} \varphi(k) \underbrace{\left[1 + \varphi^T(k) \frac{1}{\mu} S_{k-1}^{-1} \varphi(k) \right]^{-1}}_{1 \times 1} \underbrace{\varphi^T(k) \frac{1}{\mu} S_{k-1}^{-1}}_{p \times 1}$$

Note that $[\dots]$ is a scalar $\Rightarrow [\dots]^{-1} = \frac{1}{[\dots]}$

$$\Rightarrow S_k^{-1} = \frac{1}{\mu} S_{k-1}^{-1} - \cancel{\frac{1}{\mu^2}} \frac{1}{1 + \varphi^T(k) \frac{1}{\mu} S_{k-1}^{-1} \varphi(k)} S_{k-1}^{-1} \varphi(k) \varphi^T(k) S_{k-1}^{-1} \cdot \frac{1}{\mu^2}$$

We never perform any inversion, but define $V_k \triangleq S_k^{-1}$ $V_{k-1} = S_{k-1}^{-1}$

at inst. k ,
with $\hat{\theta}^*(k-1)$
 V_{k-1} } available $\Rightarrow \begin{cases} V_k = \frac{1}{\mu} V_{k-1} \frac{1}{\mu^2} \frac{1}{1 + \varphi^T(k) \frac{1}{\mu} V_{k-1} \varphi(k)} V_{k-1} \varphi(k) \varphi^T(k) V_{k-1} \\ \underline{y}(k) = V_k \varphi(k) \quad , \quad \hat{y}(k | \hat{\theta}^*(k-1)) = \varphi^T(k) \hat{\theta}^*(k-1) \\ \hat{\theta}^*(k) = \hat{\theta}^*(k-1) + \underline{y}(k) [y(k) - \hat{y}(k | \hat{\theta}^*(k-1))] \end{cases}$

$\Rightarrow$ we never compute S_k but propagate $V_k = S_k^{-1}$

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S_0 to initialize the alg. we will start with V_0 instead of S_0 .

$$V_0 = S_0^{-1} \Rightarrow S_0 > 0 \Leftrightarrow V_0 > 0.$$

How to "start" recursion.

1. "Standard" initialization

$$S_0 = I$$

$\hat{\theta}^*(0)$ "some reasonable value" (sometimes $\hat{\theta}^*(0) = \underline{0}$)

of course at the first time instants
the estimates will not be so good.

Recall: we would want $\hat{\theta}^*(k) = \left(\sum_k W_k \Phi_k \Phi_k^T \right)^{-1} \sum_k W_k \Phi_k y_k$

with standard initialization $\hat{\theta}^*(k)$ will be wrongly computed but
as $k \rightarrow \infty$ then the algorithm converges to $\hat{\theta}^*(k)$ and also

the computed S_k converges to $\sum_k W_k \Phi_k \Phi_k^T$
recursive

② Exact initialization.

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We have $S_k \triangleq \Phi_k W_k \Phi_k^T$

So, we can wait until some time instant $\bar{k}$ s.t.

$(\Phi_{\bar{k}} W_{\bar{k}} \Phi_{\bar{k}}^T)$ is pos. defined.

$(\Phi_{\bar{k}} W_{\bar{k}} \Phi_{\bar{k}}^T) > 0 \Rightarrow (\Phi_{\bar{k}} W_{\bar{k}} \Phi_{\bar{k}}^T)^{-1}$ exists.

At instant $\bar{k}$ we can compute

$$S_{\bar{k}} = \Phi_{\bar{k}} W_{\bar{k}} \Phi_{\bar{k}}^T$$

$$\text{or } V_k = (\Phi_{\bar{k}} W_{\bar{k}} \Phi_{\bar{k}}^T)^{-1}$$

↑
once the inversion
must be performed

Using (only once)
the batch solution $\hat{\theta}^*(\bar{k}) = (\Phi_{\bar{k}} W_{\bar{k}} \Phi_{\bar{k}}^T)^{-1} \Phi_{\bar{k}} W_{\bar{k}} y_{\bar{k}}$

Then for $k > \bar{k}$ we can apply the recursive alg.

of course, to have $\Phi_{\bar{k}} W_{\bar{k}} \Phi_{\bar{k}}^T > 0 \quad \bar{k} \geq p$

in fact we want $\text{rank}(\underbrace{\Phi_k}_{p \times k} \underbrace{W_k}_{k \times k} \underbrace{\Phi_k^T}_{k \times p}) = p$

$\text{rank } M = \#$ of linearly indep. rows or columns.
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We have $\text{rank}(\Phi_k W_k \Phi_k^T) = p \Leftrightarrow \text{rank } \Phi_k = p$

Recall $\Phi_k = [\psi(1) | \psi(2) | \dots | \psi(k)]$

k columns $\Rightarrow$ to have $\text{rank } \Phi_k = p$:

$$k \geq p$$

In With the "exact initialization"

first estimate for $\hat{\theta}^*(k)$ is available at $k = \bar{k}$.

If we need ~~$\hat{\theta}(k)$~~ $\hat{\theta}(k)$ also at first instant $k < \bar{k} \Rightarrow$ standard Initialization.

	PROS	CONS
Standard init.	$\hat{\theta}(k)$ always available	first estimates are "wrong"
Exact init.	$\hat{\theta}^*(k)$ computed exactly	$\hat{\theta}^*(k)$ available only from $k = \bar{k}$

one possibility is to start with standard initial.

and change to the exact computation as $(\Phi_k^T W_k \Phi_k > 0)$

$\text{rank } \Phi_k = p.$

Possibly the system we want to model could be a NON LINEAR one.

Of course one could apply a linear model (e.g. ARX, OE, ARMAX)

use the data from the system (DATA SET) and identify its parameters.

$\Rightarrow$ we apply a linear model to a ~~not~~ NON-linear system.

Sometimes not so wrong. If nonlinearities are not that strong.

Nonlinearities are treated as disturbances and a linear approx. is used.

This is often done around some (equilibrium) point for small variations.

BUT: sometimes nonlinearities are to be considered.

$\Rightarrow$ we have to resort to a NONLINEAR model.

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We could start with a "generalization of the ARX".

ARX: $\hat{y}(k|\hat{\theta}) = \varphi^T(k) \hat{\theta}$ that comes from the Hp:

$$y(k) = \varphi^T(k) \theta + n(k)$$

$$\text{with } \varphi(k) = [y(k-1), \dots, y(k-m), u(k), \dots, u(k-m)]^T$$

$\uparrow \qquad \qquad \qquad \uparrow$
"-" can be omitted.

Qualitatively $y(k) = \text{lin. combination of } \varphi(k) + n(k)$

Passing to a nonlinear case we can assume that ~~the~~ some nonlinear function $\gamma(\cdot)$ exists so that

$$y(k) = \gamma[\varphi(k)] + n(k)$$

$\Rightarrow$ the identification problem here is to find the

"unknown" function $\gamma(\cdot)$

$$\hat{\gamma}^*(\cdot) = \arg \min_{\hat{\gamma}(\cdot)} J_N(\hat{\gamma})$$

$\uparrow$ functional.